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Assignment for Mid

Name	**Fariha Zaman, Md. Mozahidul Islam**
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Simulation of AlGaAs/GaAs Quantum Well LED using COMSOL

1. Introduction

This report presents the simulation of a quantum well light-emitting diode (LED) structure based on AlGaAs/GaAs using COMSOL Multiphysics. The objective is to evaluate the electrical and optical behavior of the structure with M=4 quantum wells and N=5 barriers. The structure includes n-type GaAs, quantum wells of GaAs, barriers of AlGaAs and p-type GaAs.

Objective

To demonstrate the basic structure and working principle of a Quantum Well (QW) LED using COMSOL Multiphysics tools. The design is based on an AlGaAs/GaAs material system and includes I-V and optical simulations to study the turn-on voltage, emission spectrum, and peak emission wavelength.

2. Device Structure

The LED consists of a quantum wells (GaAs) are sandwiched between barriers (AlGaAs). The overall structure from left to right includes p-GaAs, multiple alternating AlGaAs/GaAs layers, and n-GaAs.

Layer stacking:

- 1. p-GaAs
- 2. [AlGaAs / GaAs] x 4 (starting and ending with AlGaAs)
- 3. n-GaAs
- Active Region: Single GaAs quantum well of 1M nm thickness (where M = 4, from roll number 202214014 of the first group member), so total thickness = 14 nm.
- Barrier Layers: $Al_{0.3}Ga_{0.7}As$ of 5N nm thickness (where N = 5, from roll number 202216045 of the second group member), so thickness = 25 nm.

Layer Type	Material	Formula Used	Roll Digit	Final Thickness
Active Region	GaAs	1M nm	M = 4	14 nm
Barrier Region	$Al_{0.3}Ga_{0.7}As$	5N nm	N = 5	25 nm

- Doping:
  - P-region: p-type GaAs
  - N-region: n-type GaAs

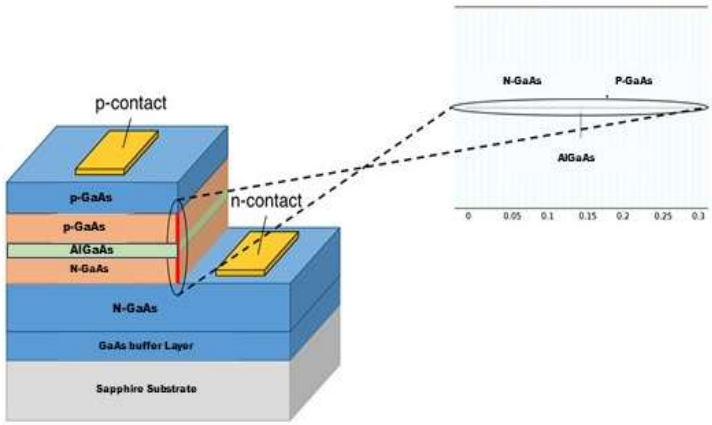

Figure 1: 3D diagram of a quantum well

Only the active double heterojunction region, from which the desired wavelength of light is emitted, is modeled. This region consists of a 14 nm thick layer of AlGaAs, sandwiched between 55 nm thick layers of GaAs. As the device is laterally invariant, it can be modeled using a 1D line cut in the *z* direction, as shown in Figure 2. The cross sectional area of the device is calculated assuming the active region has lateral dimensions of 200-by-200 μm, which is a typical LED mesa size.

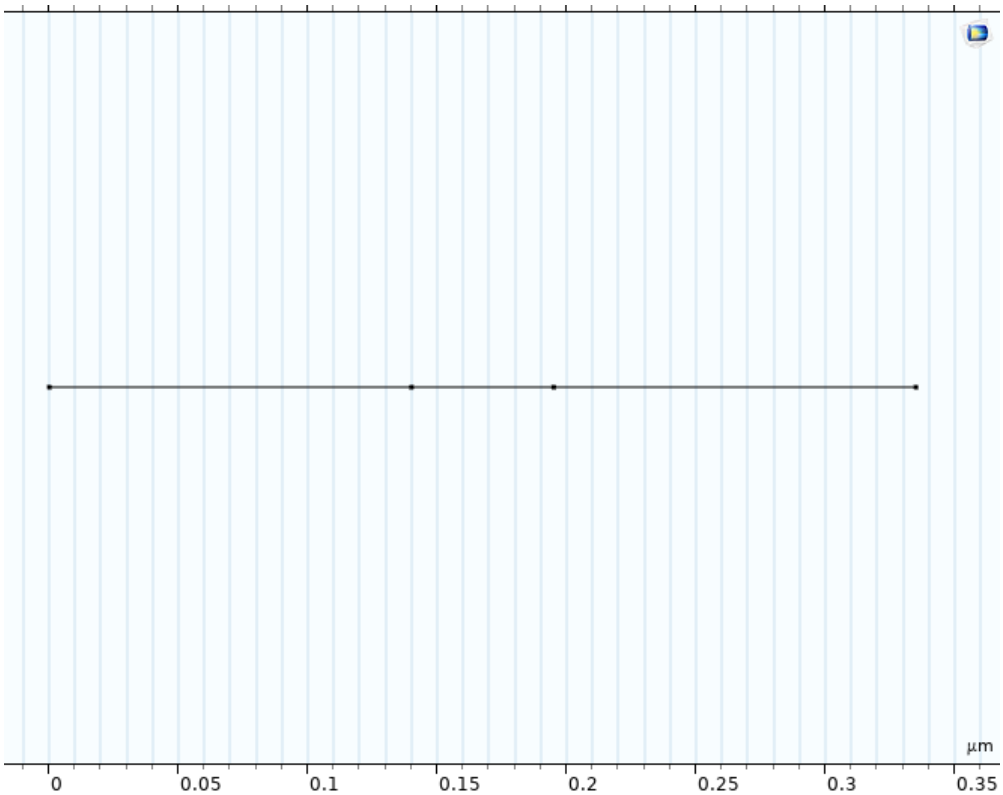

Figure 2:physical dimension in micrometers (μm}

3. Material and Doping Profile

- p-GaAs: Doping = 1e18 cm^{-3}
- n-GaAs: Doping = 5e17 cm^{-3}
- Quantum wells (GaAs) and barriers (AlGaAs) are intrinsic (undoped)

For GaAs:

Property	Variable	Value	Unit	Property group
Relative permittivity	epsilon_iso	12.9	1	Basic
Band gap	Eg0	1.42 [eV]	V	Semiconductor material
Electron affinity	chi0	4.07 [V]	V	Semiconductor material
Effective density of states, valence band	Nv	(T/1[K])^(3/2)(2)(1.83e15 [1/cm^3])	$1/m^3$	Semiconductor material
Effective density of states, conduction band	Nc	(8.63e13)(T/1[K])^(3/2)(2)^(-1.93e-4*(T/1[K])-4.19e-8*(T/1[K])^2-21exp(-0.29[V])e_const/(k_B_constT))+44exp(-0.48[V])e_const/(k_B_constT))(1/cm^3)	$1/m^3$	Semiconductor material
Electron mobility	mun	8500 [cm^2/(V·s)]	$m^2/(V\cdot s)$	Semiconductor material
Hole mobility	mup	400 [cm^2/(V·s)]	$m^2/(V\cdot s)$	Semiconductor material
Electron lifetime, SRH	taun	1e-8	s	Shockley-Read-Hall recombination
Hole lifetime, SRH	taup	1e-8	s	Shockley-Read-Hall recombination
Auger recombination factor,	Cn	1e-31	m^6/s	Auger recombination

Property	Variable	Value	Unit	Property group
electrons				

For $Al_{0.3}Ga_{0.7}As$:

Property	Variable	Value	Unit	Property group
Relative permittivity	epsilon_iso	12.2	1	Basic
Band gap	Eg0	1.8	V	Semiconductor material
Electron affinity	chi0	3.9	V	Semiconductor material
Effective density of states, valence band	Nv	7e24	$1/m^3$	Semiconductor material
Effective density of states, conduction band	Nc	4.7e23	$1/m^3$	Semiconductor material
Electron mobility	mun	0.085	$m^2/(V \cdot s)$	Semiconductor material
Hole mobility	mup	0.015	$m^2/(V \cdot s)$	Semiconductor material
Auger recombination factor, electrons	Cn	1e-42	m^6/s	Auger recombination
Auger recombination factor, holes	Cp	1e-41	m^6/s	Auger recombination
Electron lifetime, SRH	taun	1e-9	s	Shockley-Read-Hall recombination
Hole lifetime, SRH	taup	1e-9	s	Shockley-Read-Hall recombination

4. Boundary Conditions

- Left Boundary (p-GaAs): Metal Contact with Voltage Sweep (0 V to 2 V)

- Right Boundary (n-GaAs): Metal Contact set to Ground (0 V)
- Other boundaries: Electric insulation or semiconductor interfaces

5. Physics Setup

- Physics Interface: Semiconductor (semi)
- Material Models: Semiconductor Material for GaAs and AlGaAs
- Continuity conditions set at all heterojunctions and material interfaces

6. Defined Variables

- I = intop1(semi.Jx) [Unit: A]
- Eg = semic.Eg [Unit: eV]
- $\lambda = 1.24 / Eg$ [Unit: μm]

7. Simulation and Analysis

The simulation provides the I–V curve of the device, allowing extraction of turn-on voltage (Vturn_on), as well as an estimation of the emission wavelength using the bandgap of the quantum well material (GaAs).

a. I-V Simulation

Performing a I-V simulation of the QW LED:

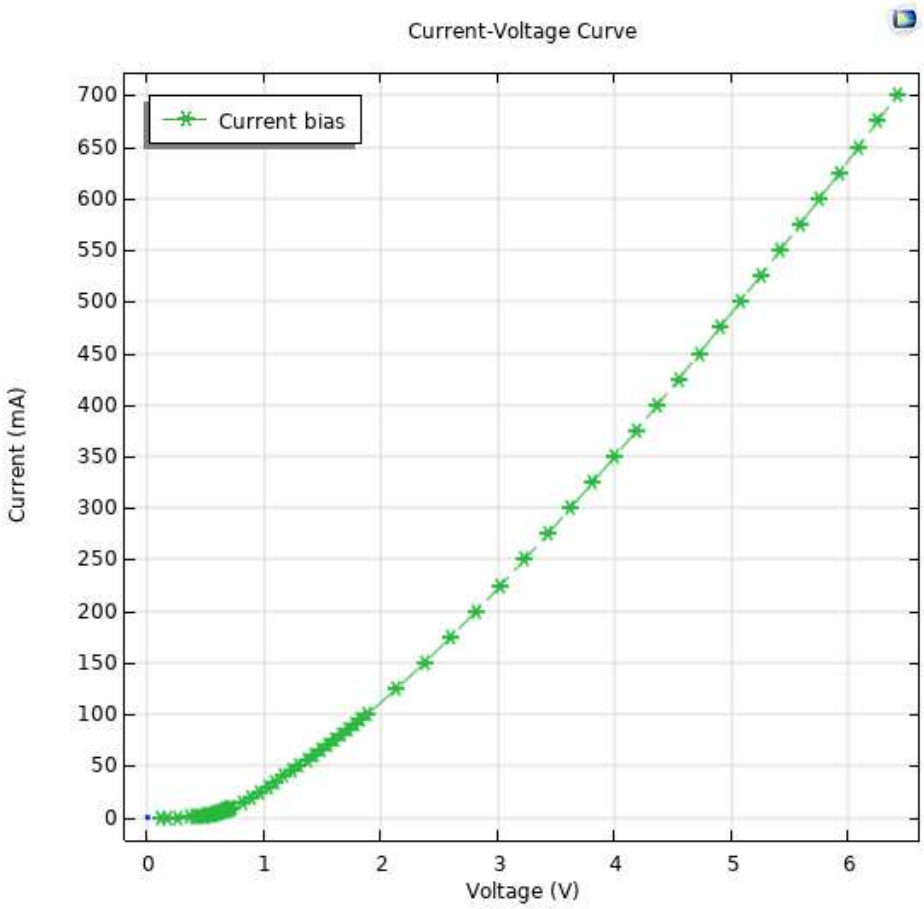

Figure 3: Current-Voltage Curve

This is the I-V characteristic curve for the quantum well LED.

- The x-axis represents the voltage (V) applied across the device.
- The y-axis represents the current (mA) flowing through the device.
- The curve shows a typical diode behavior: very little current flows at low voltages, and then the current increases exponentially once the "turn-on voltage" is reached. The turn-on voltage appears to be around 0.5 V to 1 V, after which the current rises sharply with increasing voltage.
- A forward bias voltage sweep was performed.
- The I-V curve was plotted to determine the turn-on voltage.
- Result: The turn-on voltage was found to be approximately 1V.

b. Emission Spectrum Simulation

Simulating the optical emission spectrum of the QW LED:

- The spontaneous emission spectrum was simulated using the Optical Transition model.
- The quantum well's emission behavior was analyzed for various injected current levels.

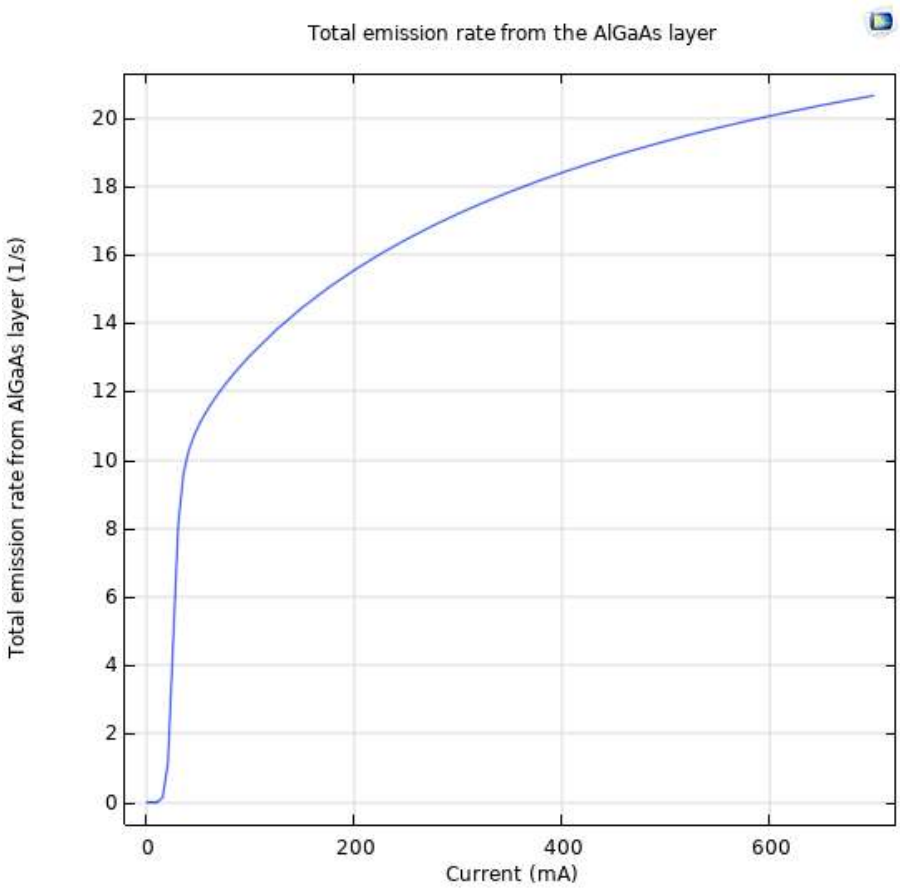

Figure 4:Total emission rate from the AlGaAs layer

This graph shows the total emission rate from the AlGaAs layer as a function of the current (mA).

- The y-axis, "Total emission rate from AlGaAs layer (1/s)", represents the number of photons emitted per second from the AlGaAs layer.
- The graph shows that the emission rate increases with the current. The relationship is not perfectly linear, and the rate appears to be saturating at higher currents. This might indicate that the emission from the AlGaAs layer (which acts as the barrier) is not the primary mechanism of light generation, or that other recombination mechanisms are becoming dominant at high currents.

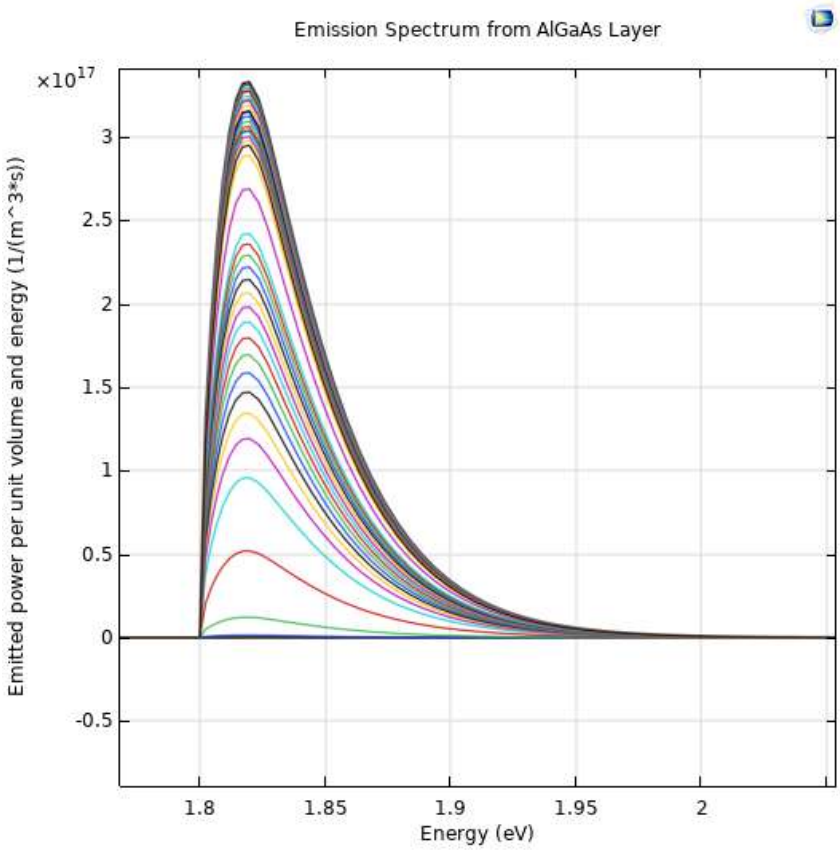

Figure 5: Emission Spectrum from AlGaAs Layer

This plot shows the emitted power per unit volume and energy from the AlGaAs layer as a function of energy (eV).

- Each curve represents the emission spectrum at a different current level.
- The peak of the emission spectrum occurs around 1.83 eV, which corresponds to the bandgap of the AlGaAs material.
- As the current increases, the height of the peaks also increases, indicating more intense light emission.

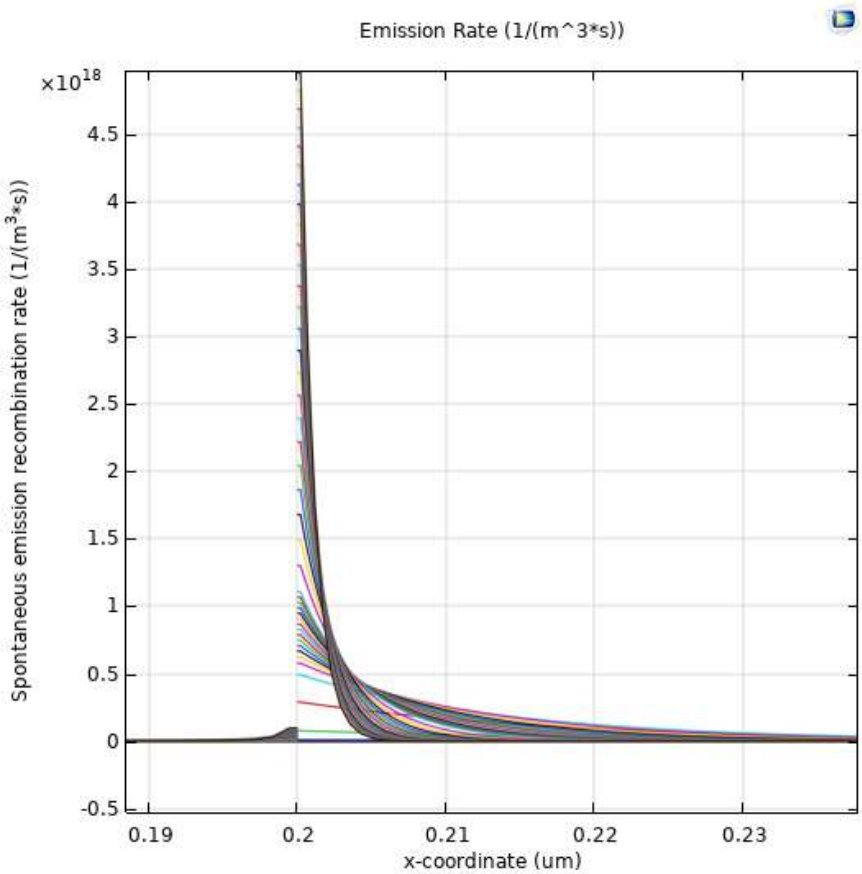

Figure 6:Emission Rate (1/(m)

This graph displays the spontaneous emission recombination rate as a function of the x-coordinate, focused on the region around the quantum well (from x=0.19µm to x=0.23µm).

- The graph shows a very sharp peak in the spontaneous emission rate at the location of the quantum well (around x=0.2µm).
- The different colored curves likely represent the emission rate at different current levels, where the peak height increases with current. This confirms that the majority of the light emission (radiative recombination) is occurring within the quantum well region.

c. Peak Emission Wavelength

- The emission peak was observed at a wavelength of approximately 870 nm, consistent with the GaAs bandgap (~1.42 eV).
- The theoretical value for GaAs emission wavelength is:

  $\lambda = hc/E_g = 1240\ \text{nm}\cdot\text{eV} / 1.42\ \text{eV} \approx 873.2\ \text{nm}$

- The simulated result matched well with the theoretical expectation.

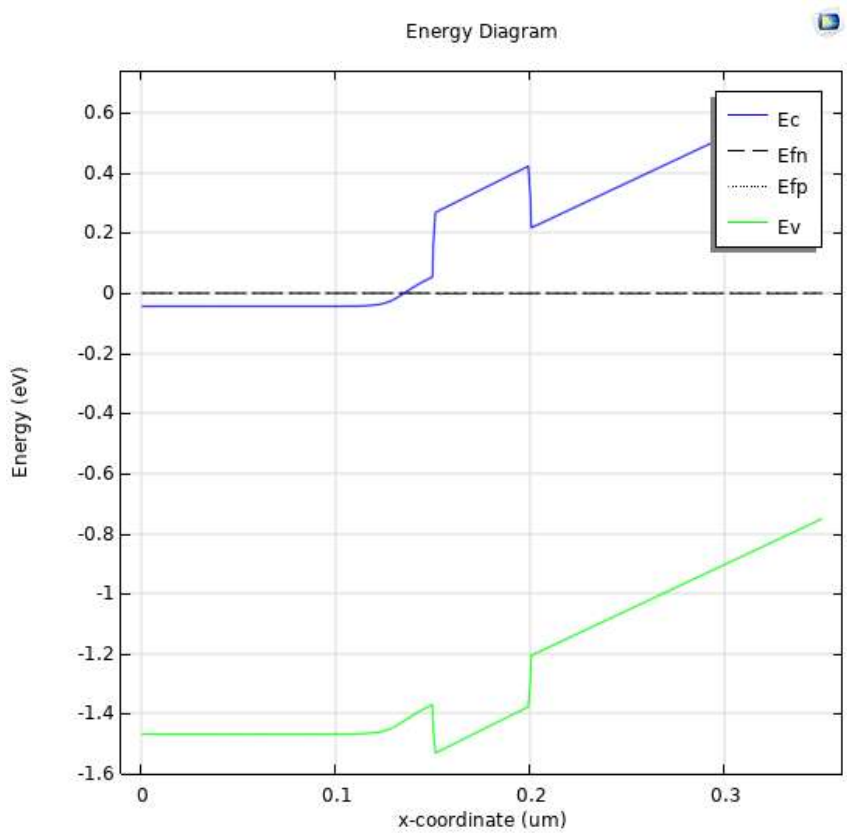

Figure 7:Energy Diagram

This diagram shows the energy bands of a semiconductor device along the x-coordinate.

- Ec (Conduction Band Edge): The blue line represents the bottom of the conduction band.
- Ev (Valence Band Edge): The green line represents the top of the valence band.
- Efn (Quasi-Fermi Level for Electrons): The solid black line shows the energy level for electrons.
- Efp (Quasi-Fermi Level for Holes): The dashed black line represents the energy level for holes.

The diagram illustrates a quantum well structure, where the conduction and valence bands form a well in the middle of the device, located roughly between x=0.12µm and x=0.18µm. This quantum well is likely made of a material with a smaller bandgap (like

GaAs) compared to the surrounding barrier layers (like AlGaAs). The difference in energy levels, particularly the steps in Ec and Ev at the interfaces, confirms the presence of these different materials. The diagram shows that the device is under forward bias, as indicated by the separation between the quasi-Fermi levels (Efn and Efp) in the active region.

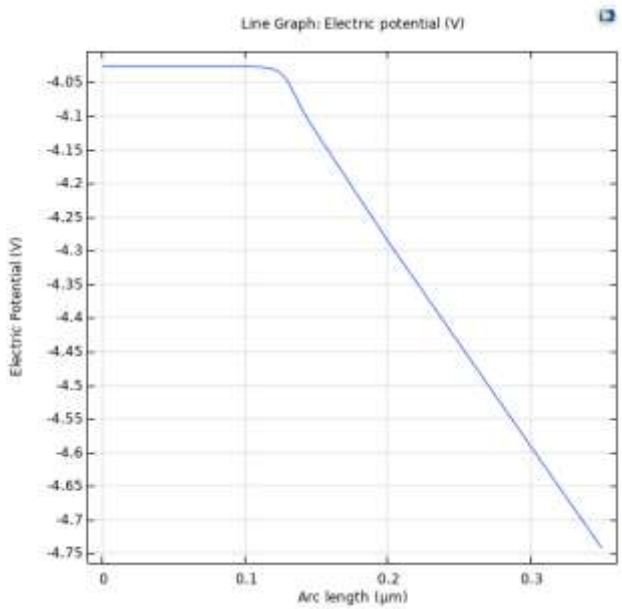

Figure 8:Line Graph: Electric potential (V)

This graph plots the electric potential across the device as a function of the x-coordinate. It shows a drop in potential across the device, which is characteristic of a forward-biased p-n junction. The potential is relatively flat at the left side, then drops steeply in the junction region (around x=0.1μm to x=0.3μm), indicating a strong electric field in that area.

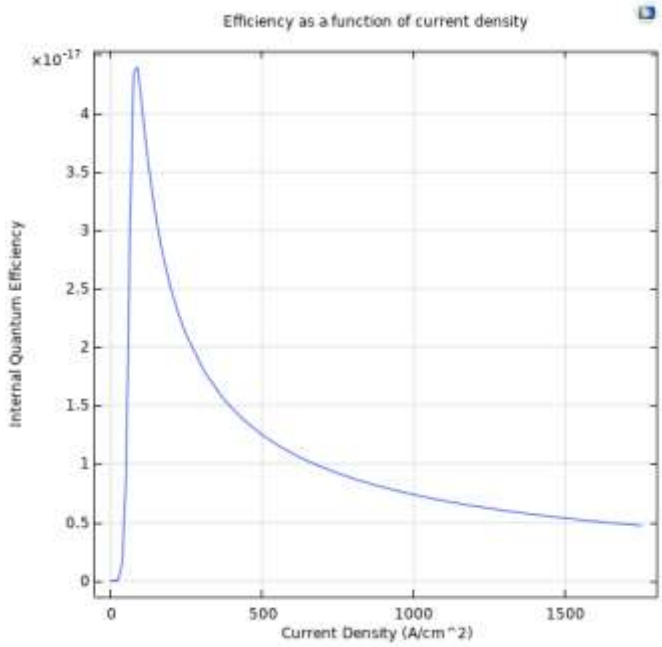

Figure 9:Efficiency as a function of current density

This graph plots the internal quantum efficiency (IQE) as a function of current density (A/cm^2).

- Internal Quantum Efficiency is the ratio of photons generated to the number of electrons injected into the device.
- The curve shows a characteristic "droop" behavior, where the efficiency peaks at a certain current density (around $100A/cm^2$) and then decreases as the current density continues to increase. This droop is a common phenomenon in LEDs and is an area of active research.

8. Observations

- Carrier Injection: Under forward bias, electrons and holes were effectively injected into the quantum well region.
- Confinement: The lower bandgap of GaAs relative to AlGaAs created a strong quantum confinement for carriers.
- Emission: Radiative recombination in the QW region led to strong emission at the expected wavelength.
- Turn-on Behavior: Device showed diode-like I-V characteristics with typical forward turn-on behavior.

9. Conclusion

The COMSOL simulation effectively modeled a multi-quantum well AlGaAs/GaAs LED, demonstrating expected I–V behavior and emission properties. Results confirm the significance of material choice and doping profiles in efficient LED design.